

Pitehr Hurtado-Cayo

Ph.D. Candidate in Industrial Engineering

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Personal Profile

Senior Data Scientist specializing in optimization, machine learning and large-scale decision systems. Experienced in leading technical initiatives for Advertising and Marketplace platforms across Latin America, designing optimization algorithms, production ML systems and scalable decision architectures that deliver measurable business impact. Ph.D. candidate in Industrial Engineering with research experience at MIT focused on Operations Research, Optimization and Machine Learning.

Industry Positions

MercadoLibre Chile

Santiago de Chile, Chile

Senior Data Scientist

Jan 2024 - Current

- Led the design and deployment of optimization systems for Advertising across Latin America, improving campaign delivery by 15% through intraday budget allocation and experimentation strategies.
- Designed scalable decision systems combining optimization and machine learning to support real-time budget allocation under latency and operational constraints.
- Partnered with Product, Backend and ML teams to drive the technical design and deployment of optimization solutions for regional advertising platforms.
- Developed and deployed production machine learning models for customer behavior analysis, improving targeting effectiveness and increasing marketing ROI by 20%.
- Designed and analyzed online experiments (A/B testing) to validate optimization strategies and support product decision-making using statistical evidence.

Previously: Data Scientist (Sep 2022 – Dec 2023)

Projects

Advertising Budget Optimization Platform (Mercado Libre)

- Designed the optimization formulation and distributed architecture for intraday advertising budget allocation across LATAM.
- Led experimentation strategy through online A/B testing.

Dynamic Bid Pricing Optimization

Designed a dynamic pricing optimization framework that adjusted bid prices based on real-time campaign performance, maximizing delivery efficiency while adapting to changing marketplace conditions.

Education

Pontificia Universidad Católica de Valparaíso (PUCV)

Valparaíso, Chile

Ph.D. Candidate in Industrial Engineering

2026 - Current

- Research focus: Operations Research, Mathematical Optimization, Decision Systems and Machine Learning.

M.Sc. in Industrial Engineering (Summa Cum Laude)

2021

Thesis: Branch-and-Price Algorithms for Large-Scale Optimization.

B.Sc. in Industrial Engineering

2020

GPA: 6.2 / 7.0 (highest GPA out of 55 students).

Massachusetts Institute of Technology (MIT)

Boston, USA

Visiting Researcher

2024

- Conducted research on large-scale optimization and decision systems.
- Collaborated with researchers from the MIT Center for Transportation and Logistics.

Teaching Experience

Capstone Project Advisor, School of Industrial Engineering

Valparaíso, Chile

Advised capstone projects for Bachelor's Program Industrial Engineering

Aug 2024 - Current

Guide undergraduate capstone teams in developing industrial engineering solutions aligned with academic and industry standards.

Teaching Assistant

Valparaíso, Chile

Faculty of Engineering, PUCV

Teaching assistant for undergraduate and graduate courses in Operations Research, Optimization and Data Analytics.

Honors & Awards

School of Industrial Engineering, PUCV 2016 - 2020

Member of the Honor List, which comprises the top ten GPAs of the whole School of Industrial Engineering (over 800 students).

PUCV 2015 - 2020

Obtained the PUCV outstanding entrance scholarship, awarded to students with an average PAES greater than 780 (out of 850).

Additional Information

Programming	Cplex, Gurobi, C++, Python (Pandas, PyTorch, NumPy, Scikit-learn. etc.), Java, SQL.
Languages	Spanish — Native, English — Professional working proficiency
Conference Presentation	Presented the main results of the Master’s thesis at the XIII Chilean Conference on Operations Research.
Other	Participant of the Work and Travel program at Stowe Mountain Resort in VT, USA (Dec 2019 - Mar 2020).